Tasks on Terraform-06

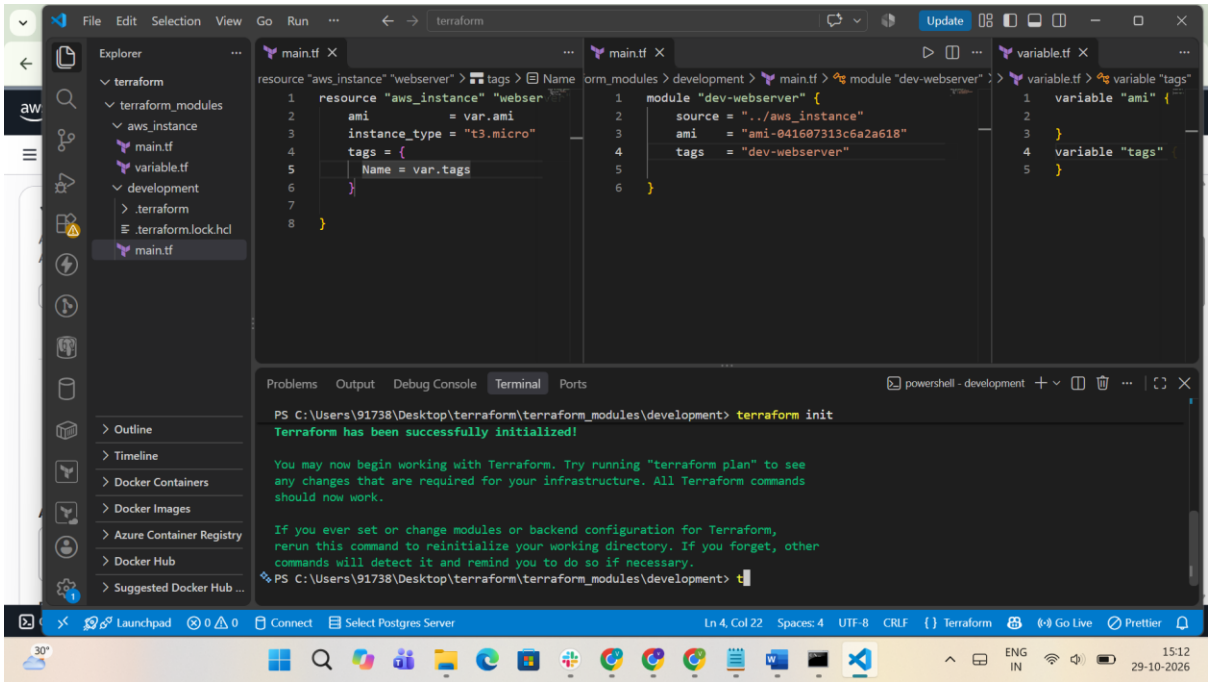
1). Watch the Terraform-06 video.

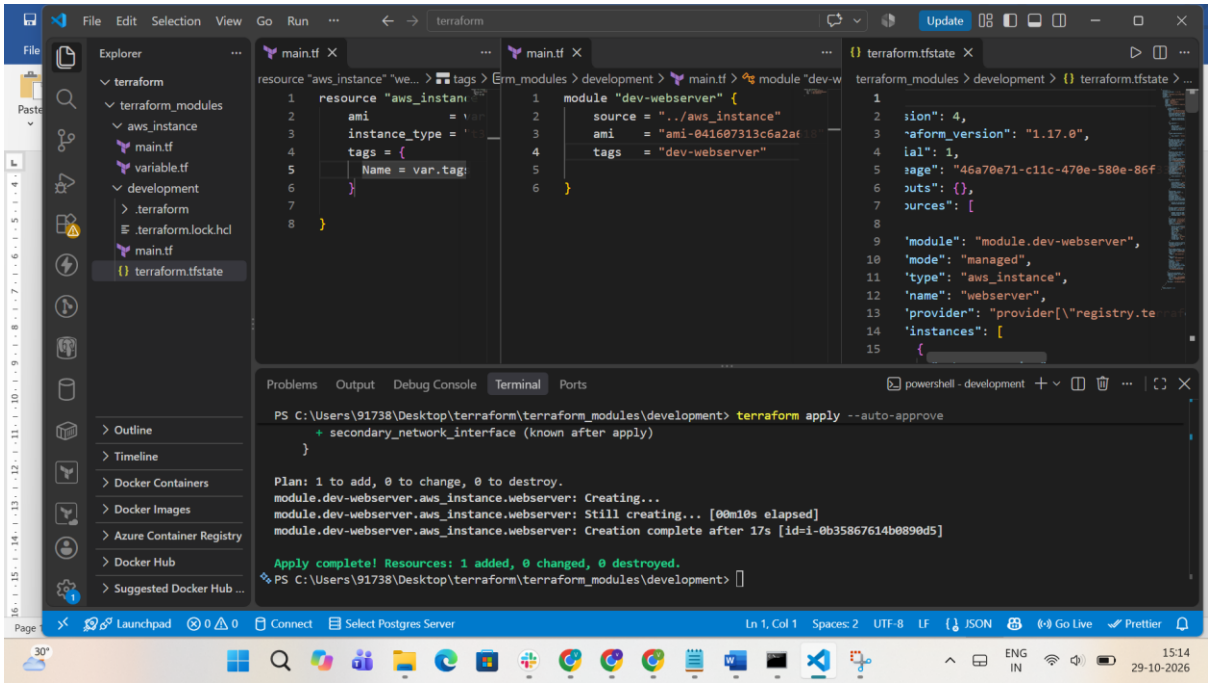
Terraform-06 video completed.

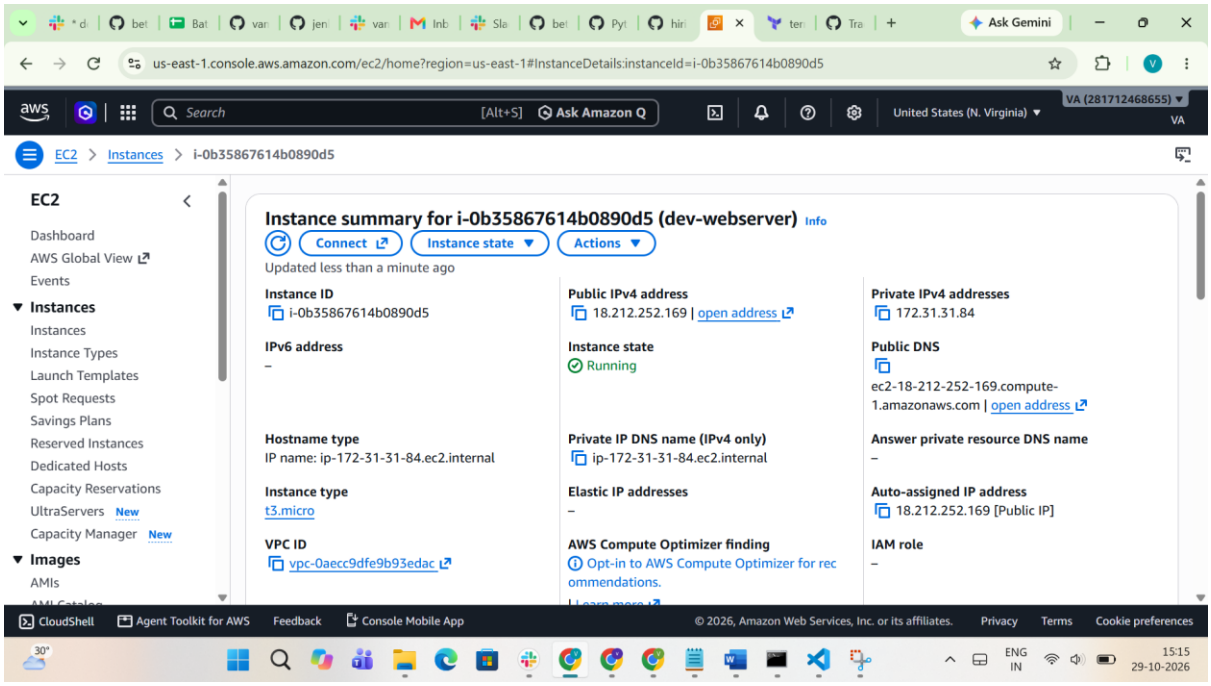
2). Execute the script shown in the video.

Implementations steps:

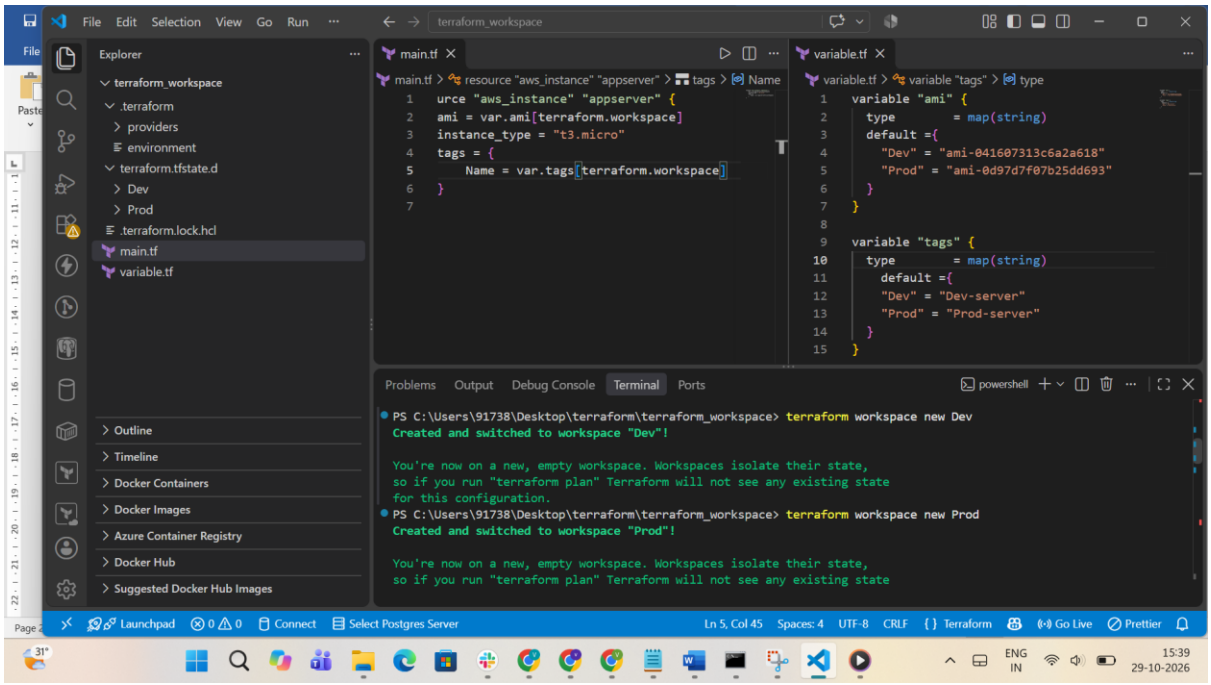
1.Terraform modules:

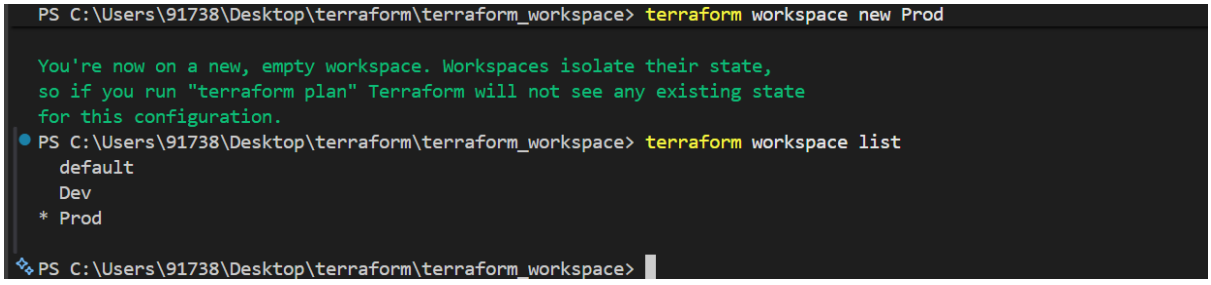


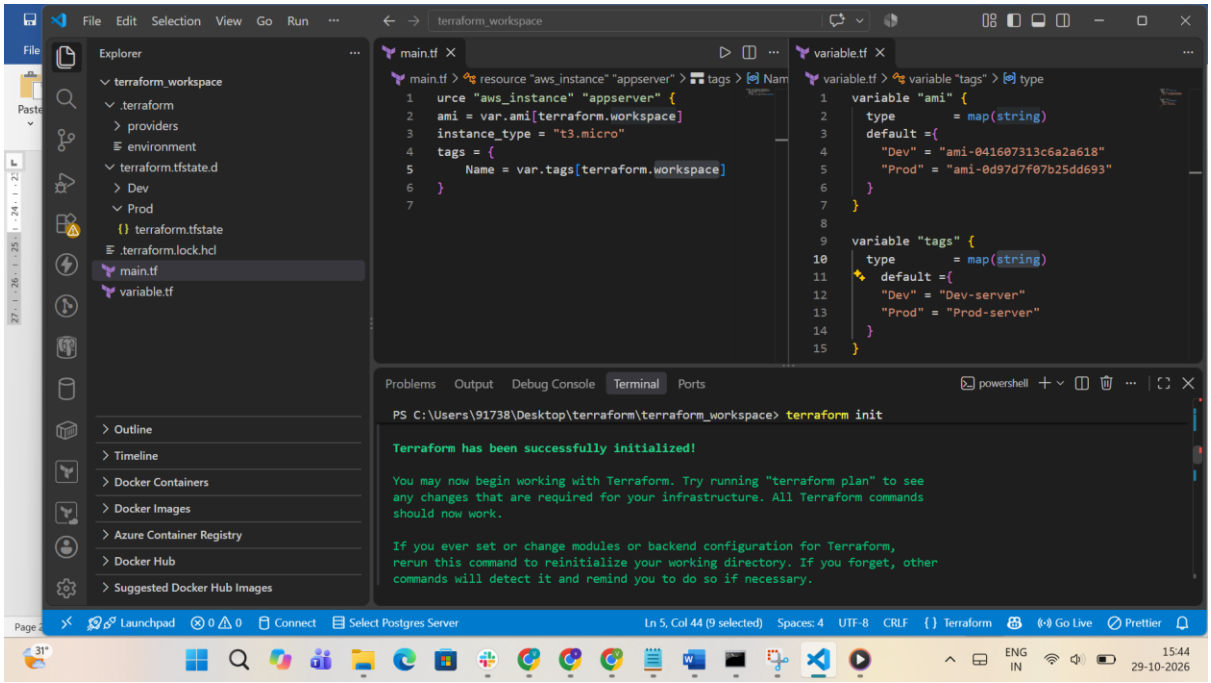


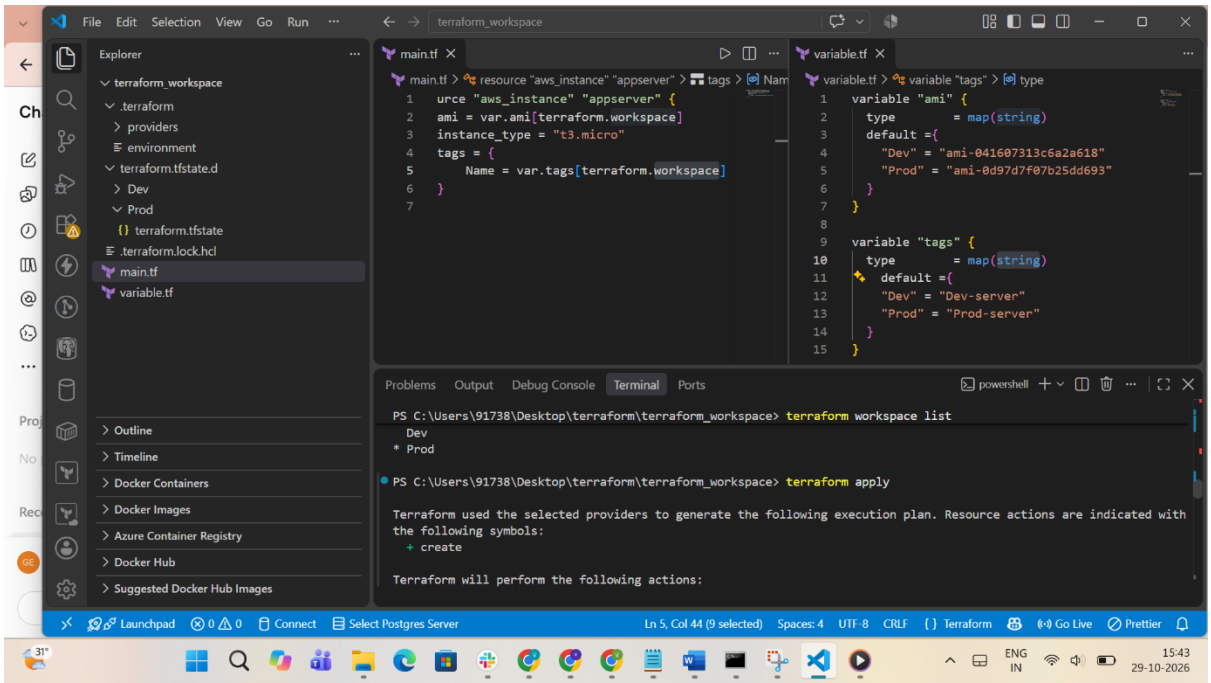


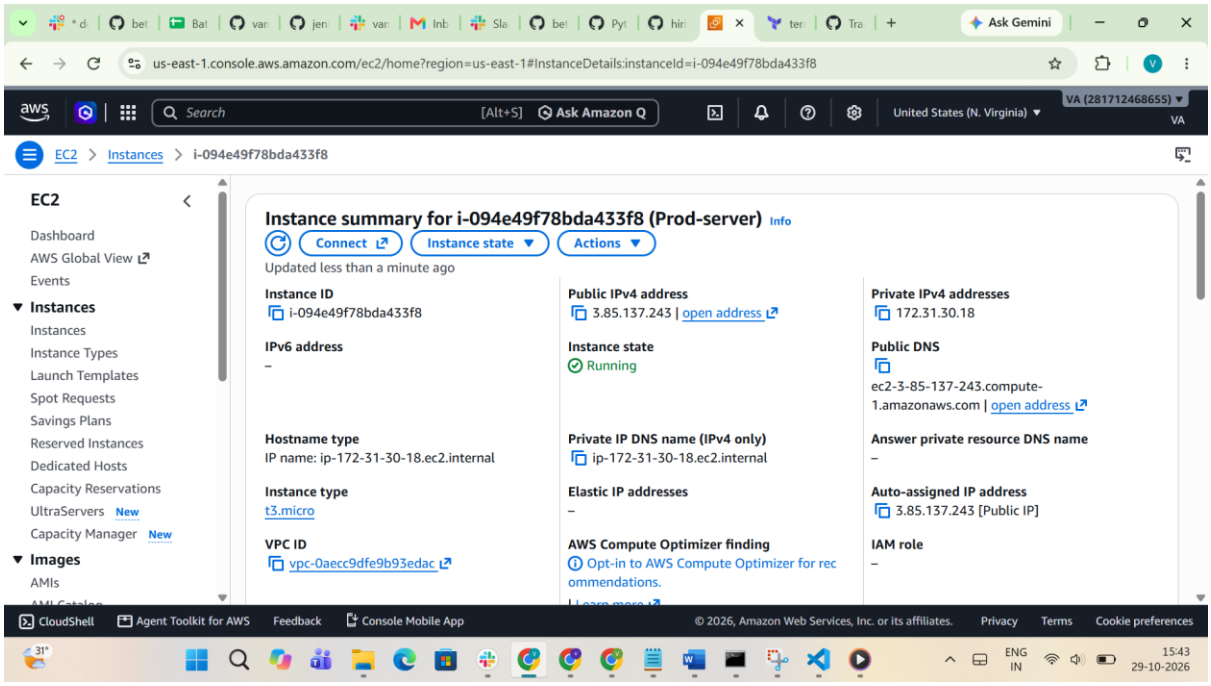
2.Terraform workspace:

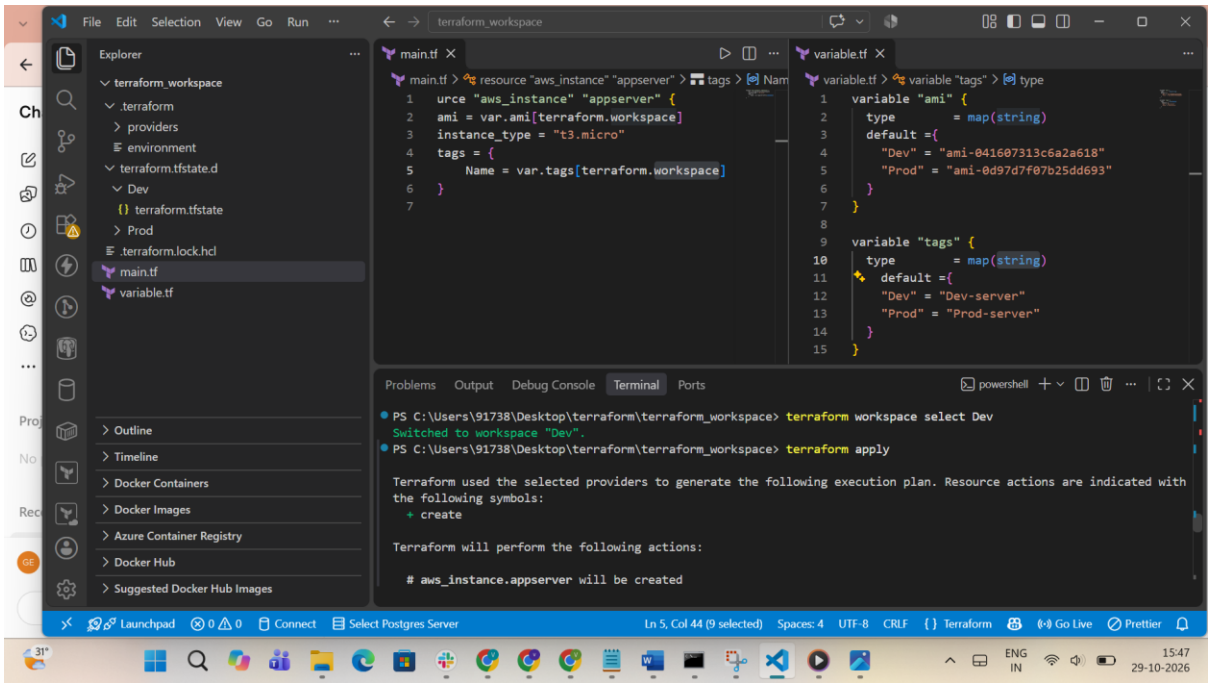


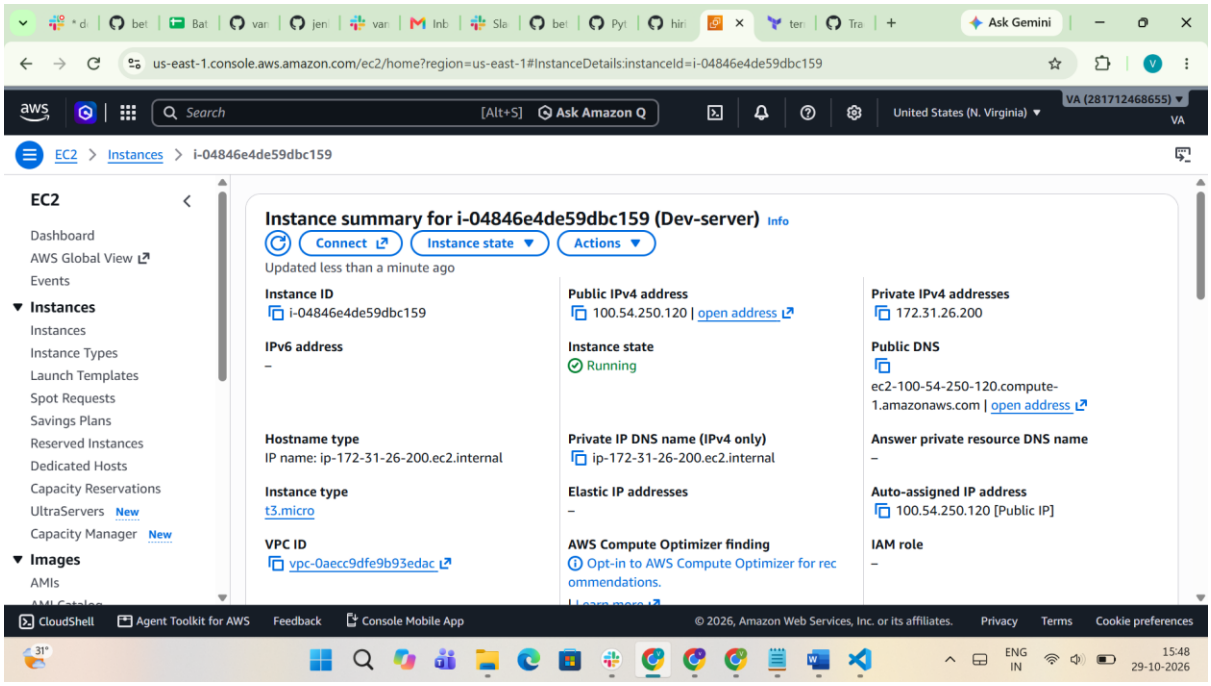



File Edit Selection View Go Run
terraform_workspace
Explorer
terraform_workspace
.terraform
providers
environment
terraform.tfstate.d
Dev
Prod
terraform.tfstate
.terraform.lock.hcl
main.tf
variable.tf
main.tf
variable.tf
urce "aws_instance" "appserver" {
ami = var.ami[terraform.workspace]
instance_type = "t3.micro"
tags = {
Name = var.tags[terraform.workspace]
}
variable "ami" {
type = map(string)
default ={
"Dev" = "ami-041607313c6a2a618"
"Prod" = "ami-0d97d7f07b25dd693"
}
}
variable "tags" {
type = map(string)
default ={
"Dev" = "Dev-server"
"Prod" = "Prod-server"
}
}
Problems Output Debug Console Terminal Ports
powershell
PS C:\Users\91738\Desktop\terraform\terraform_workspace> terraform init
Terraform has been successfully initialized!
You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.
If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
Outline
Timeline
Docker Containers
Docker Images
Azure Container Registry
Docker Hub
Suggested Docker Hub Images
Launchpad
Connect
Select Postgres Server
Ln 5, Col 44 (9 selected)
Spaces: 4
UTF-8
CRLF
Terraform
Go Live
Prettier
ENG
IN
15:44
29-10-2026


File Edit Selection View Go Run
terraform_workspace
Explorer
terraform_workspace
.terraform
providers
environment
terraform.tfstate.d
Dev
Prod
terraform.tfstate
.terraform.lock.hcl
main.tf
variable.tf
main.tf
variable.tf
urce "aws_instance" "appserver" {
ami = var.ami[terraform.workspace]
instance_type = "t3.micro"
tags = {
Name = var.tags[terraform.workspace]
}
variable "ami" {
type = map(string)
default ={
"Dev" = "ami-041607313c6a2a618"
"Prod" = "ami-0d97d7f07b25dd693"
}
}
variable "tags" {
type = map(string)
default ={
"Dev" = "Dev-server"
"Prod" = "Prod-server"
}
}
Problems Output Debug Console Terminal Ports
powershell
PS C:\Users\91738\Desktop\terraform\terraform_workspace> terraform workspace list
Dev
* Prod
PS C:\Users\91738\Desktop\terraform\terraform_workspace> terraform apply
Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with
the following symbols:
+ create
Terraform will perform the following actions:
Outline
Timeline
Docker Containers
Docker Images
Azure Container Registry
Docker Hub
Suggested Docker Hub Images
Launchpad
Connect
Select Postgres Server
Ln 5, Col 44 (9 selected)
Spaces: 4
UTF-8
CRLF
Terraform
Go Live
Prettier
ENG
IN
15:43
29-10-2026


us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#InstanceDetails:instanceId=i-094e49f78bda433f8
Search
[Alt+S]
Ask Amazon Q
United States (N. Virginia)
VA (281712468655)
EC2 > Instances > i-094e49f78bda433f8
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
Instance summary for i-094e49f78bda433f8 (Prod-server) Info
Connect
Instance state
Actions
Updated less than a minute ago
Instance ID
i-094e49f78bda433f8
IPv6 address
–
Hostname type
IP name: ip-172-31-30-18.ec2.internal
Instance type
t3.micro
VPC ID
vpc-0aecc9dfe9b93edac
Public IPv4 address
3.85.137.243 | open address
Instance state
Running
Private IP DNS name (IPv4 only)
ip-172-31-30-18.ec2.internal
Elastic IP addresses
–
AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations.
Private IPv4 addresses
172.31.30.18
Public DNS
ec2-3-85-137-243.compute-1.amazonaws.com | open address
Answer private resource DNS name
–
Auto-assigned IP address
3.85.137.243 [Public IP]
IAM role
–
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
15:43
29-10-2026


File Edit Selection View Go Run
terraform_workspace
Explorer
terraform_workspace
.terraform
providers
environment
terraform.tfstate.d
Dev
terraform.tfstate
Prod
.terraform.lock.hcl
main.tf
variable.tf
main.tf
resource "aws_instance" "appserver" > tags
urce "aws_instance" "appserver" {
ami = var.ami[terraform.workspace]
instance_type = "t3.micro"
tags = {
Name = var.tags[terraform.workspace]
}
variable.tf
variable "tags" > type
variable "ami" {
type = map(string)
default ={
"Dev" = "ami-041607313c6a2a618"
"Prod" = "ami-0d97d7f07b25dd693"
}
}
variable "tags" {
type = map(string)
default ={
"Dev" = "Dev-server"
"Prod" = "Prod-server"
}
}
Problems
Output
Debug Console
Terminal
Ports
powershell
PS C:\Users\91738\Desktop\terraform\terraform_workspace> terraform workspace select Dev
Switched to workspace "Dev".
PS C:\Users\91738\Desktop\terraform\terraform_workspace> terraform apply
Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create
Terraform will perform the following actions:
aws_instance.appserver will be created
Outline
Timeline
Docker Containers
Docker Images
Azure Container Registry
Docker Hub
Suggested Docker Hub Images
Launchpad
Connect
Select Postgres Server
Ln 5, Col 44 (9 selected)
Spaces: 4
UTF-8
CRLF
Terraform
Go Live
Prettier
15:47
29-10-2026



3). Provision EC2, S3, and VPC using Terraform modules.

Objective

To create reusable Terraform modules for provisioning an EC2 instance, an S3 bucket, and a VPC in AWS.

Implementation steps:

1. Installed Terraform on the local machine and configured AWS credentials.
2. Created a project directory named terraform-modules.
3. Created a root configuration using main.tf, variables.tf, and outputs.tf.
4. Created separate directories for the vpc, ec2, and s3 modules.
5. Created the VPC module to provision a VPC and subnet.
6. Created the EC2 module to provision an EC2 instance in the subnet created by the VPC module.
7. Created the S3 module to provision an S3 bucket with public access blocked.
8. Called the three modules from the root main.tf file and passed the required values between modules.
9. Executed terraform init to initialize Terraform and download the AWS provider.
10. Executed terraform validate to validate the configuration.
11. Executed terraform plan to review the resources that would be created.

12. Executed terraform apply and confirmed the deployment.
13. Verified the VPC, subnet, EC2 instance, and S3 bucket in the AWS Management Console.

Project folder structure

```
terraform-modules/
|
├── main.tf
├── variables.tf
├── outputs.tf
|
└── modules/
    ├── aws_vpc/
    |   ├── main.tf
    |   └── outputs.tf
    |
    ├── aws_ec2/
    |   ├── main.tf
    |   └── outputs.tf
    |
    └── aws_s3/
        └── main.tf
```

Code 1: VPC module

File: modules/aws_vpc/main.tf

This module creates a VPC, a public subnet, an Internet Gateway, and a route that allows internet traffic to reach the public subnet.

```
resource "aws_vpc" "main" {
  cidr_block           = "10.0.0.0/16"
  enable_dns_support   = true
  enable_dns_hostnames = true
```

```
  tags = {
    Name = "terraform-vpc"
  }
}

resource "aws_subnet" "public" {
  vpc_id                  = aws_vpc.main.id
  cidr_block              = "10.0.1.0/24"
  map_public_ip_on_launch = true

  tags = {
    Name = "terraform-public-subnet"
  }
}

resource "aws_internet_gateway" "main" {
  vpc_id = aws_vpc.main.id

  tags = {
    Name = "terraform-igw"
  }
}

resource "aws_route_table" "public" {
  vpc_id = aws_vpc.main.id

  route {
    cidr_block = "0.0.0.0/0"
```

```
    gateway_id = aws_internet_gateway.main.id
  }

  tags = {
    Name = "terraform-public-route-table"
  }
}

resource "aws_route_table_association" "public" {
  subnet_id      = aws_subnet.public.id
  route_table_id = aws_route_table.public.id
}
```

File: modules/aws_vpc/outputs.tf

```
output "vpc_id" {
  value = aws_vpc.main.id
}

output "subnet_id" {
  value = aws_subnet.public.id
}
```

Code 2: EC2 module

File: modules/aws_ec2/main.tf

This module creates an EC2 instance inside the VPC subnet and installs Apache.

```
data "aws_ami" "amazon_linux" {
  most_recent = true
  owners      = ["amazon"]
```

```
  filter {
    name   = "name"
    values = ["al2023-ami-*-x86_64"]
  }

  filter {
    name   = "architecture"
    values = ["x86_64"]
  }
}

resource "aws_security_group" "web" {
  name   = "terraform-module-http-sg"
  vpc_id = var.vpc_id

  ingress {
    from_port   = 80
    to_port     = 80
    protocol    = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }

  egress {
    from_port   = 0
    to_port     = 0
    protocol    = "-1"
    cidr_blocks = ["0.0.0.0/0"]
  }
}
```

```
resource "aws_instance" "web" {
  ami                    = data.aws_ami.amazon_linux.id
  instance_type          = var.instance_type
  subnet_id              = var.subnet_id
  vpc_security_group_ids = [aws_security_group.web.id]
  associate_public_ip_address = true

  user_data = <<-EOF
    #!/bin/bash
    dnf install -y httpd
    systemctl enable --now httpd
    echo "<h1>EC2 Created Using Terraform Modules</h1>" > /var/www/html/index.html
  EOF

  tags = {
    Name = var.instance_name
  }
}
```

File: modules/aws_ec2/variables.tf

```
variable "vpc_id" {
  type = string
}

variable "subnet_id" {
  type = string
}
```

```
variable "instance_type" {
  type    = string
  default = "t3.micro"
}

variable "instance_name" {
  type = string
}
```

File: modules/ec2/outputs.tf

```
output "instance_id" {
  value = aws_instance.web.id
}

output "public_ip" {
  value = aws_instance.web.public_ip
}
```

Code 3: S3 module

File: modules/aws_s3/main.tf

```
variable "bucket_name" {
  type = string
}

resource "aws_s3_bucket" "main" {
  bucket = var.bucket_name

  tags = {
    Name = "terraform-module-bucket"
```

```
  }
}

resource "aws_s3_bucket_public_access_block" "main" {
  bucket = aws_s3_bucket.main.id

  block_public_acls       = true
  block_public_policy     = true
  ignore_public_acls      = true
  restrict_public_buckets = true
}

output "s3_bucket_name" {
  value = module.s3.s3_bucket_name
}
```

Code 4: Root configuration

File: main.tf

The root configuration calls the three modules and passes the VPC and subnet IDs to the EC2 module.

```
terraform {
  required_providers {
    aws = {
      source  = "hashicorp/aws"
      version = "~> 6.70.0"
    }
  }
}

provider "aws" {
```

```
  region = "us-east-1"
}

module "vpc" {
  source = "./modules/aws_vpc"
}

module "ec2" {
  source = "./modules/aws_ec2"

  vpc_id       = module.vpc.vpc_id
  subnet_id    = module.vpc.subnet_id
  instance_type = "t3.micro"
  instance_name = "module-web-server"
}

module "s3" {
  source      = "./modules/aws_s3"
  bucket_name = var.bucket_name
}
```

File: variables.tf in the root folder

```
variable "bucket_name" {
  description = "Globally unique S3 bucket name"
  type        = string
}
```

File: outputs.tf in the root folder

```
output "vpc_id" {
```

```
  value = module.vpc.vpc_id
}

output "ec2_instance_id" {
  value = module.ec2.instance_id
}

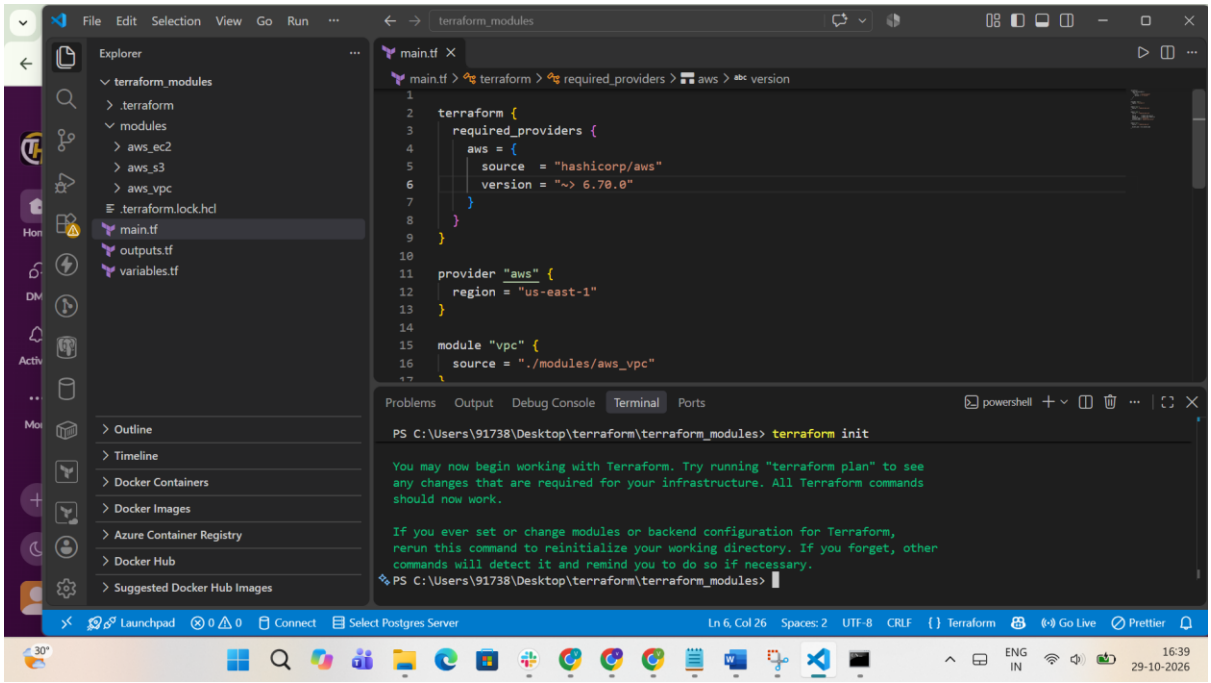
output "ec2_public_ip" {
  value = module.ec2.public_ip
}

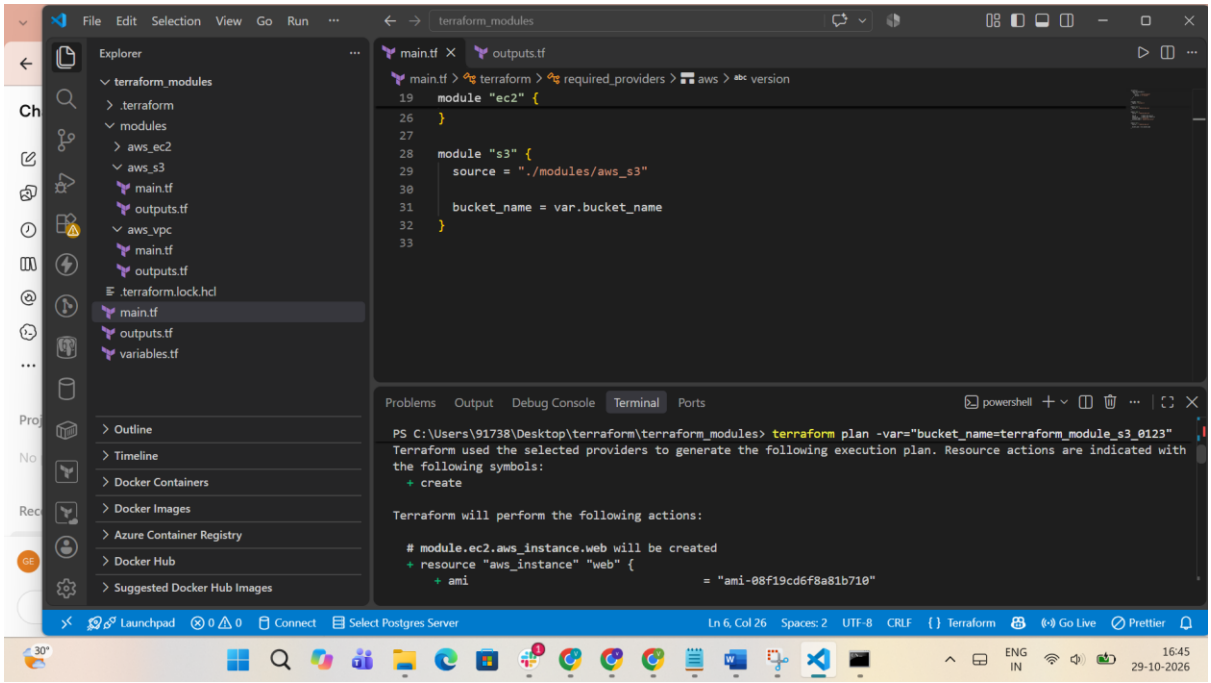
output "s3_bucket_name" {
  value = module.s3.bucket_name
}
```

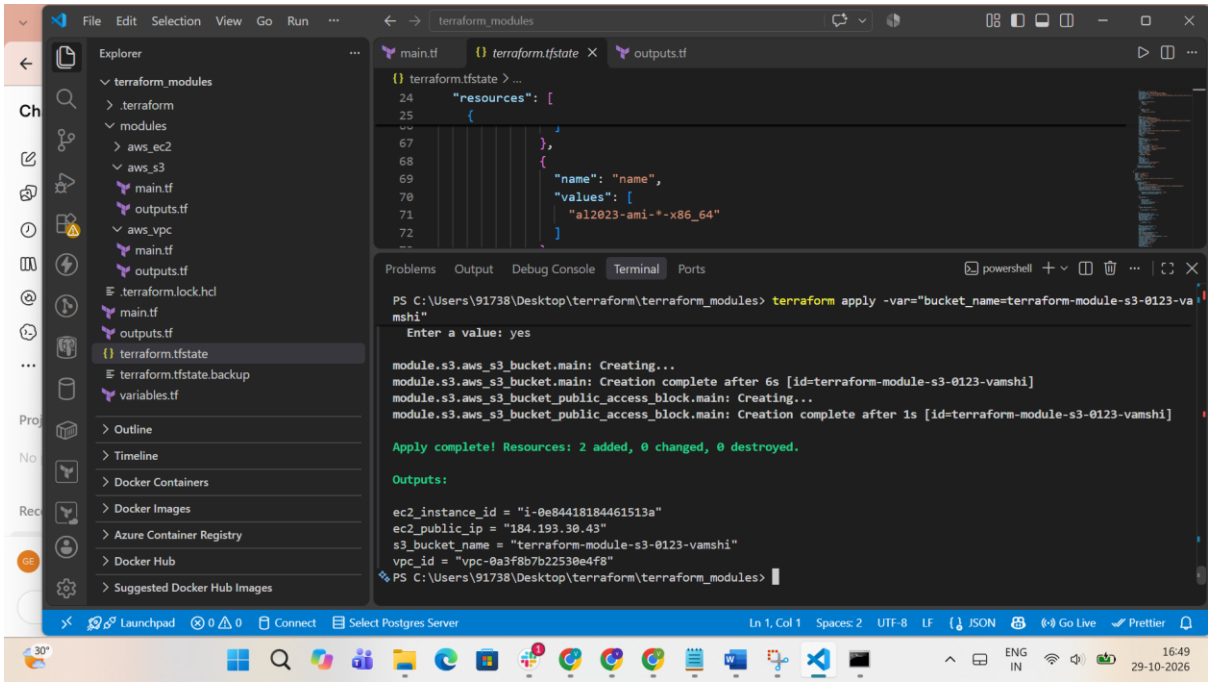
Commands executed

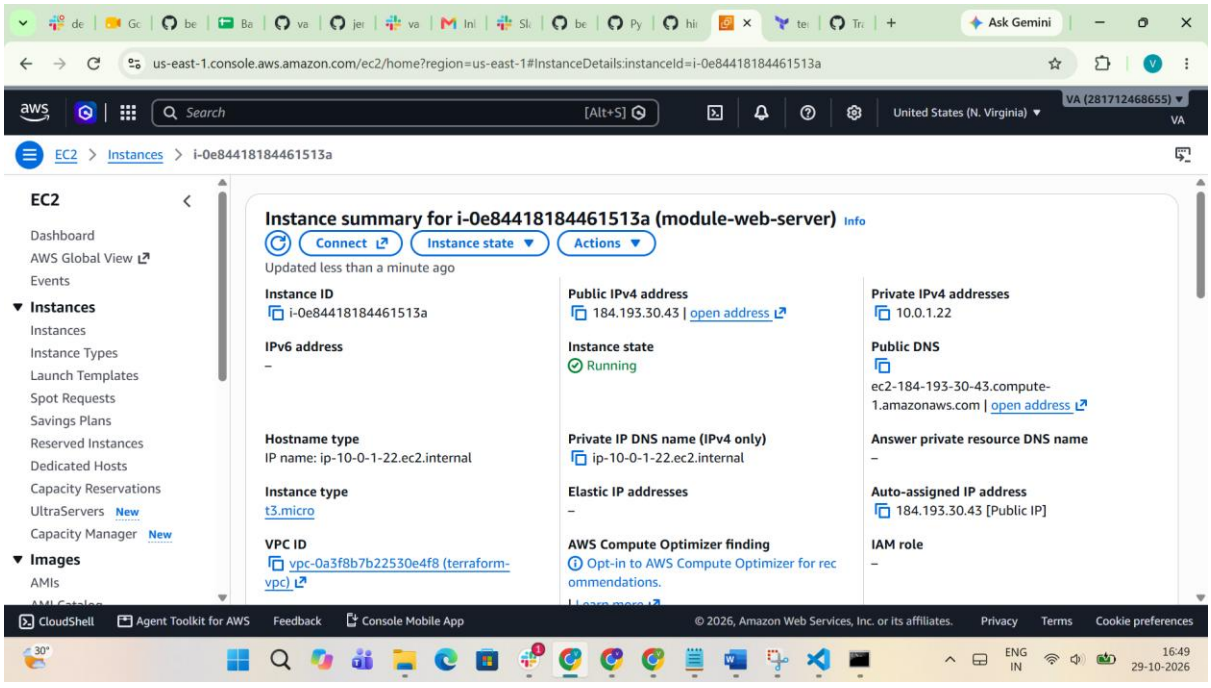
First, provide a globally unique S3 bucket name:

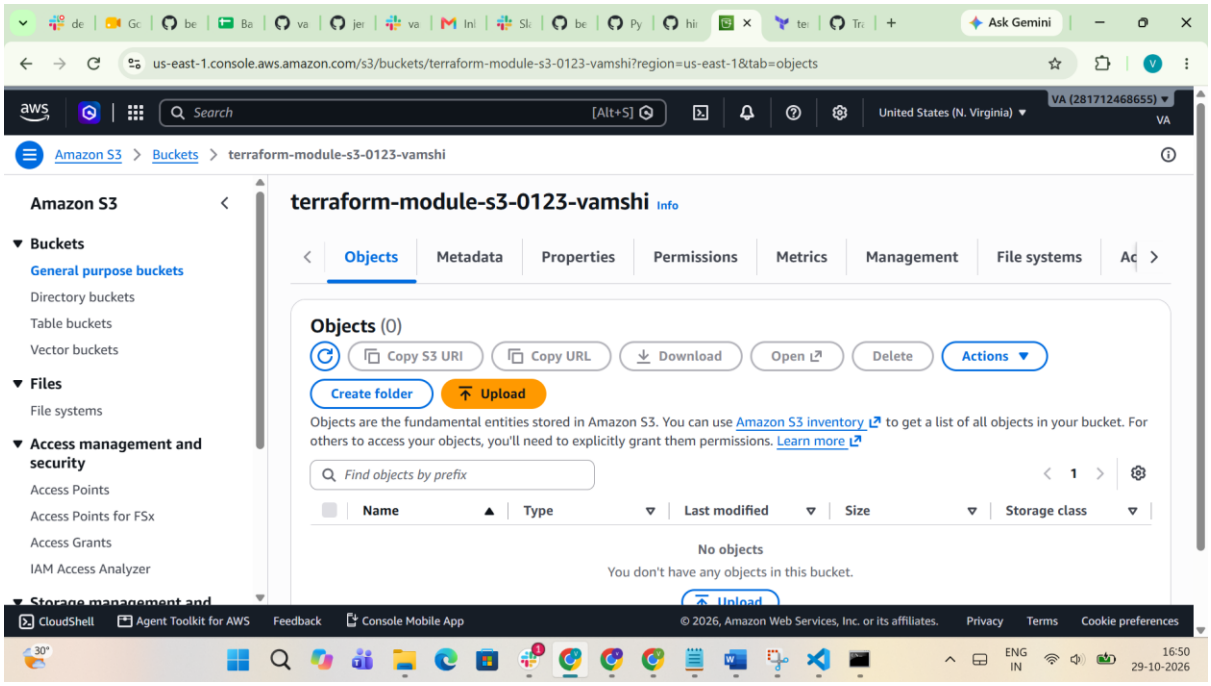
```
terraform init
terraform validate
terraform plan -var="bucket_name=your-unique-bucket-name-2026"
terraform apply -var="bucket_name=your-unique-bucket-name-2026"
```

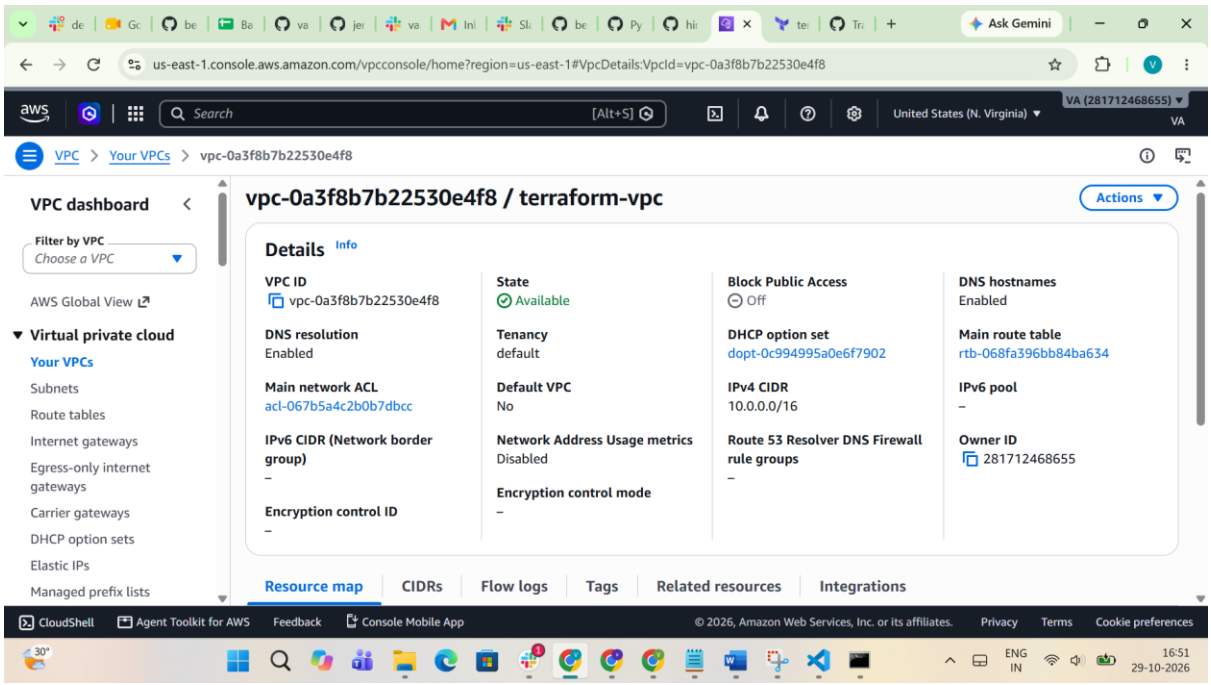

File Edit Selection View Go Run
terraform_modules
Explorer
terraform_modules
.terraform
modules
aws_ec2
aws_s3
aws_vpc
.terraform.lock.hcl
main.tf
outputs.tf
variables.tf
main.tf
main.tf > terraform > required_providers > aws > version
terraform {
required_providers {
aws = {
source = "hashicorp/aws"
version = "~> 6.70.0"
}
}
}
provider "aws" {
region = "us-east-1"
}
module "vpc" {
source = "./modules/aws_vpc"
Problems Output Debug Console Terminal Ports
powershell
PS C:\Users\91738\Desktop\terraform\terraform_modules> terraform init
You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.
If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
PS C:\Users\91738\Desktop\terraform\terraform_modules>
Outline
Timeline
Docker Containers
Docker Images
Azure Container Registry
Docker Hub
Suggested Docker Hub Images
Launchpad
Connect
Select Postgres Server
Ln 6, Col 26
Spaces: 2
UTF-8
CRLF
Terraform
Go Live
Prettier
ENG
IN
16:39
29-10-2026

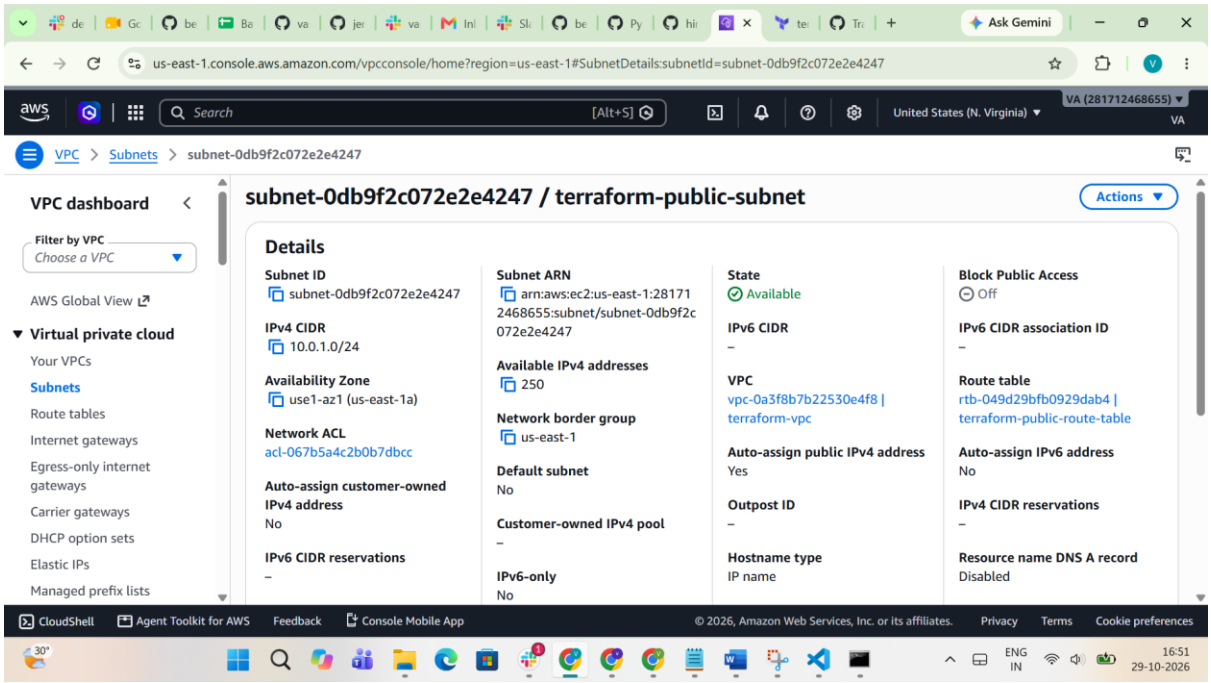

File Edit Selection View Go Run
terraform_modules
Explorer
terraform_modules
.terraform
modules
aws_ec2
aws_s3
main.tf
outputs.tf
aws_vpc
main.tf
outputs.tf
.terraform.lock.hcl
main.tf
outputs.tf
variables.tf
main.tf
outputs.tf
main.tf > terraform > required_providers > aws > version
module "ec2" {
}
module "s3" {
source = "./modules/aws_s3"
bucket_name = var.bucket_name
}
Problems Output Debug Console Terminal Ports
powershell
PS C:\Users\91738\Desktop\terraform\terraform_modules> terraform plan -var="bucket_name=terraform_module_s3_0123"
Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with
the following symbols:
+ create
Terraform will perform the following actions:
module.ec2.aws_instance.web will be created
+ resource "aws_instance" "web" {
+ ami = "ami-08f19cd6f8a81b710"
Outline
Timeline
Docker Containers
Docker Images
Azure Container Registry
Docker Hub
Suggested Docker Hub Images
Launchpad
Connect
Select Postgres Server
Ln 6, Col 26
Spaces: 2
UTF-8
CRLF
Terraform
Go Live
Prettier
ENG
IN
16:45
29-10-2026

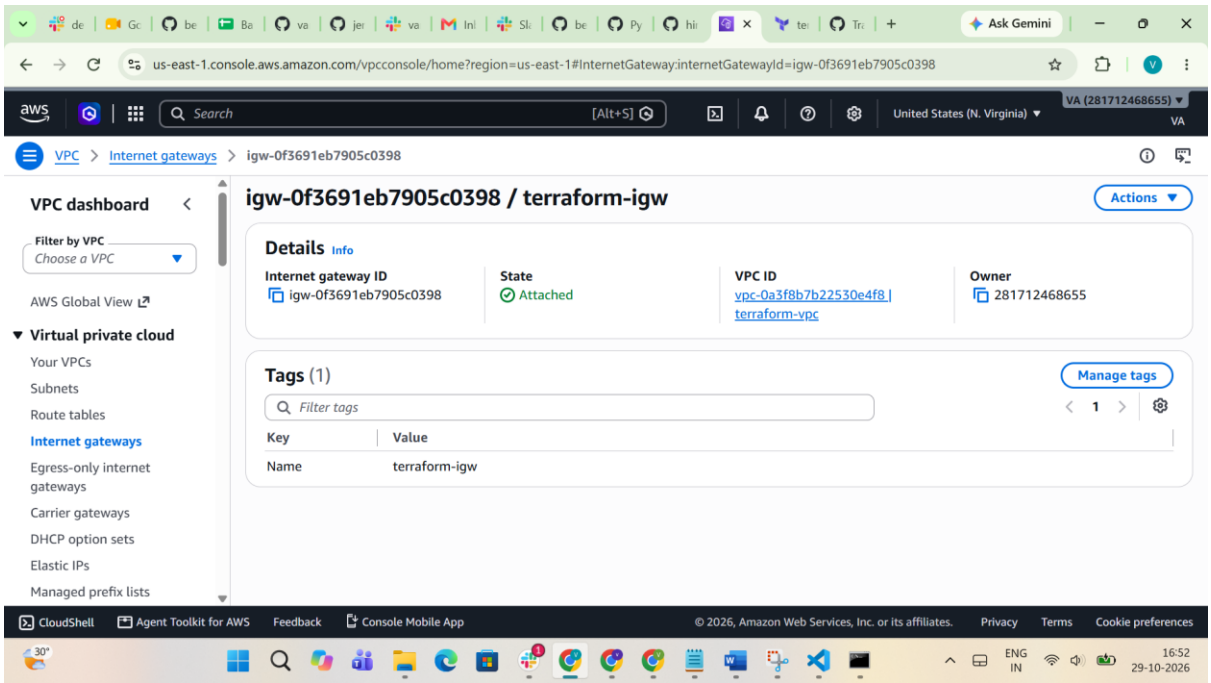

File Edit Selection View Go Run
terraform_modules
Explorer
terraform_modules
.terraform
modules
aws_ec2
aws_s3
main.tf
outputs.tf
aws_vpc
main.tf
outputs.tf
.terraform.lock.hcl
main.tf
outputs.tf
terraform.tfstate
terraform.tfstate.backup
variables.tf
Outline
Timeline
Docker Containers
Docker Images
Azure Container Registry
Docker Hub
Suggested Docker Hub Images
main.tf
terraform.tfstate
outputs.tf
terraform.tfstate > ...
24 "resources": [
25 {
67 },
68 {
69 "name": "name",
70 "values": [
71 "al2023-ami-*-x86_64"
72]
Problems Output Debug Console Terminal Ports
powershell
PS C:\Users\91738\Desktop\terraform\terraform_modules> terraform apply -var="bucket_name=terraform-module-s3-0123-vamshi"
Enter a value: yes
module.s3.aws_s3_bucket.main: Creating...
module.s3.aws_s3_bucket.main: Creation complete after 6s [id=terraform-module-s3-0123-vamshi]
module.s3.aws_s3_bucket_public_access_block.main: Creating...
module.s3.aws_s3_bucket_public_access_block.main: Creation complete after 1s [id=terraform-module-s3-0123-vamshi]
Apply complete! Resources: 2 added, 0 changed, 0 destroyed.
Outputs:
ec2_instance_id = "i-0e84418184461513a"
ec2_public_ip = "184.193.30.43"
s3_bucket_name = "terraform-module-s3-0123-vamshi"
vpc_id = "vpc-0a3f8b7b22530e4f8"
PS C:\Users\91738\Desktop\terraform\terraform_modules>
Launchpad
Connect
Select Postgres Server
Ln 1, Col 1 Spaces: 2 UTF-8 LF JSON Go Live Prettier
ENG IN
16:49 29-10-2026

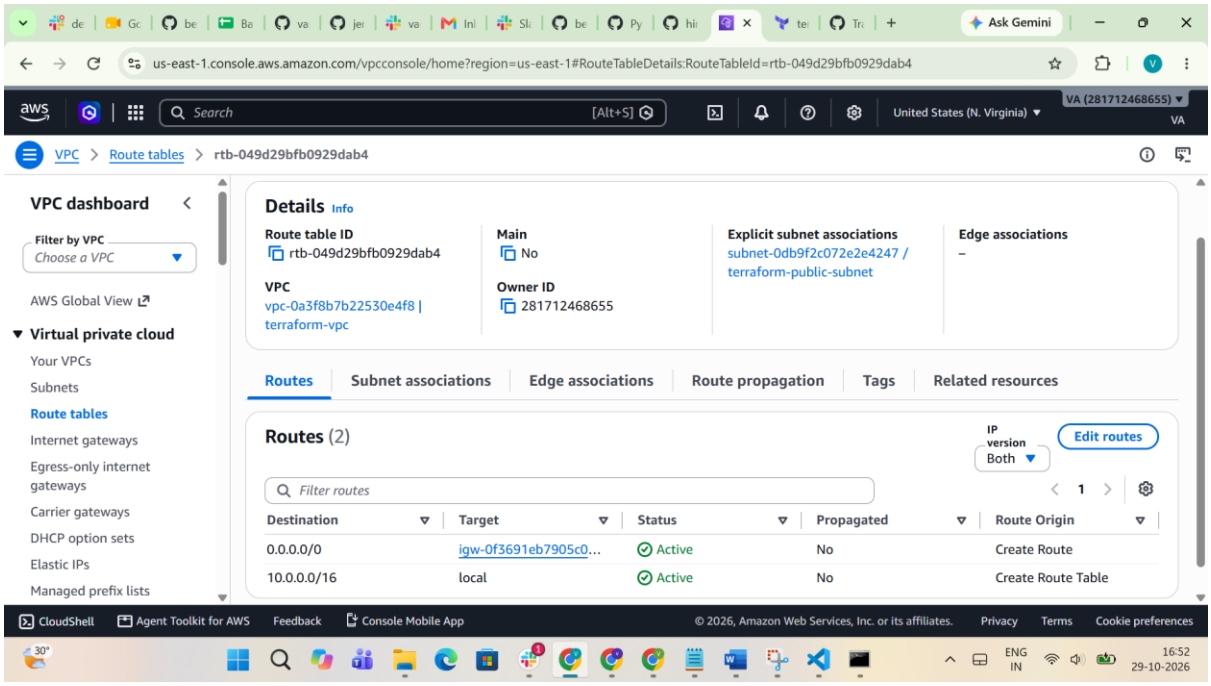

Ask Gemini
us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#InstanceDetails:instanceId=i-0e84418184461513a
Search [Alt+S]
United States (N. Virginia)
VA (281712468655)
VA
EC2 > Instances > i-0e84418184461513a
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
Instance summary for i-0e84418184461513a (module-web-server) Info
Connect
Instance state
Actions
Updated less than a minute ago
Instance ID
i-0e84418184461513a
IPv6 address
–
Hostname type
IP name: ip-10-0-1-22.ec2.internal
Instance type
t3.micro
VPC ID
vpc-0a3f8b7b22530e4f8 (terraform-vpc)
Public IPv4 address
184.193.30.43 | open address
Instance state
Running
Private IP DNS name (IPv4 only)
ip-10-0-1-22.ec2.internal
Elastic IP addresses
–
AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations.
Private IPv4 addresses
10.0.1.22
Public DNS
ec2-184-193-30-43.compute-1.amazonaws.com | open address
Answer private resource DNS name
–
Auto-assigned IP address
184.193.30.43 [Public IP]
IAM role
–
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ENG IN
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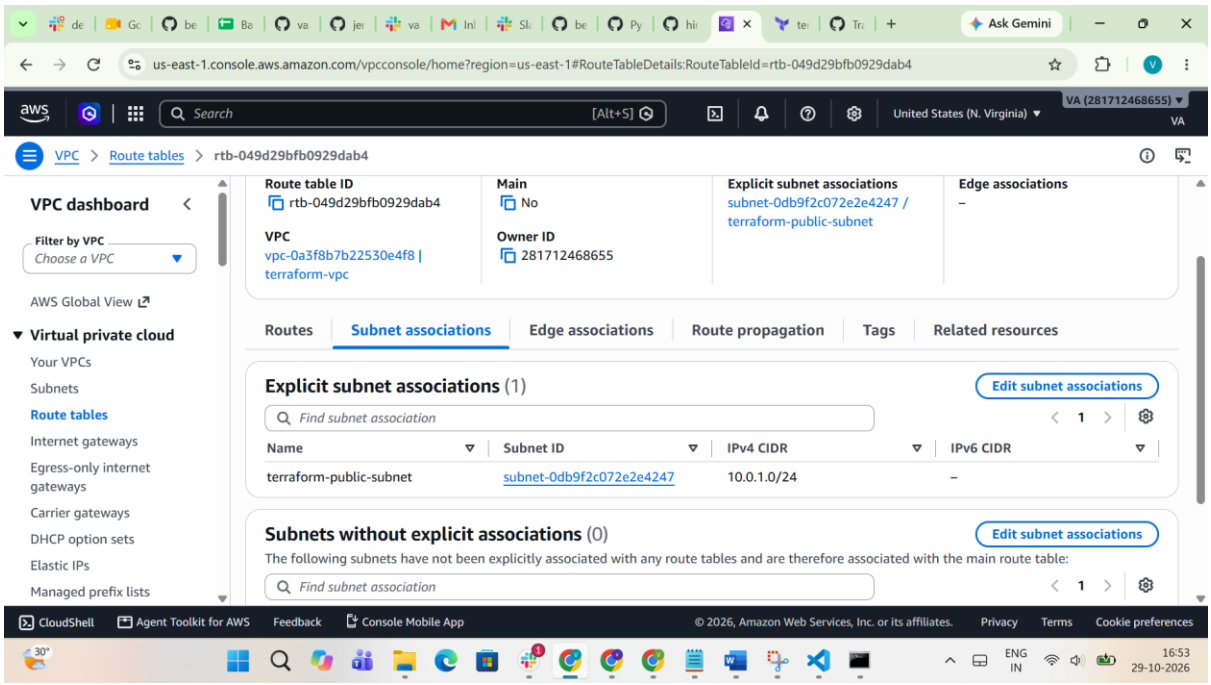

us-east-1.console.aws.amazon.com/s3/buckets/terraform-module-s3-0123-vamshi?region=us-east-1&tab=objects
Search
[Alt+S]
United States (N. Virginia)
VA (281712468655)
Amazon S3 > Buckets > terraform-module-s3-0123-vamshi
Amazon S3
Buckets
General purpose buckets
Directory buckets
Table buckets
Vector buckets
Files
File systems
Access management and security
Access Points
Access Points for FSx
Access Grants
IAM Access Analyzer
terraform-module-s3-0123-vamshi Info
Objects
Metadata
Properties
Permissions
Metrics
Management
File systems
Objects (0)
Copy S3 URI
Copy URL
Download
Open
Delete
Actions
Create folder
Upload
Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more
Find objects by prefix
Name
Type
Last modified
Size
Storage class
No objects
You don't have any objects in this bucket.
Upload
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us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#VpcDetails:VpcId=vpc-0a3f8b7b22530e4f8
Search
[Alt+S]
United States (N. Virginia)
VA (281712468655)
VPC > Your VPCs > vpc-0a3f8b7b22530e4f8
VPC dashboard
Filter by VPC
Choose a VPC
AWS Global View
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
Carrier gateways
DHCP option sets
Elastic IPs
Managed prefix lists
vpc-0a3f8b7b22530e4f8 / terraform-vpc
Actions
Details Info
VPC ID
vpc-0a3f8b7b22530e4f8
State
Available
Block Public Access
Off
DNS hostnames
Enabled
DNS resolution
Enabled
Tenancy
default
DHCP option set
dopt-0c994995a0e6f7902
Main route table
rtb-068fa396bb84ba634
Main network ACL
acl-067b5a4c2b0b7dbcc
Default VPC
No
IPv4 CIDR
10.0.0.0/16
IPv6 pool
–
IPv6 CIDR (Network border group)
–
Network Address Usage metrics
Disabled
Route 53 Resolver DNS Firewall rule groups
–
Owner ID
281712468655
Encryption control ID
–
Encryption control mode
–
Resource map
CIDRs
Flow logs
Tags
Related resources
Integrations
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us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#SubnetDetails:subnetId=subnet-0db9f2c072e2e4247
Search
United States (N. Virginia)
VPC > Subnets > subnet-0db9f2c072e2e4247
VPC dashboard
Filter by VPC
Choose a VPC
AWS Global View
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
Carrier gateways
DHCP option sets
Elastic IPs
Managed prefix lists
subnet-0db9f2c072e2e4247 / terraform-public-subnet
Actions
Details
Subnet ID
subnet-0db9f2c072e2e4247
IPv4 CIDR
10.0.1.0/24
Availability Zone
use1-az1 (us-east-1a)
Network ACL
acl-067b5a4c2b0b7dbcc
Auto-assign customer-owned IPv4 address
No
IPv6 CIDR reservations
–
Subnet ARN
arn:aws:ec2:us-east-1:281712468655:subnet/subnet-0db9f2c072e2e4247
Available IPv4 addresses
250
Network border group
us-east-1
Default subnet
No
Customer-owned IPv4 pool
–
IPv6-only
No
State
Available
IPv6 CIDR
–
VPC
vpc-0a3f8b7b22530e4f8 | terraform-vpc
Auto-assign public IPv4 address
Yes
Outpost ID
–
Hostname type
IP name
Block Public Access
Off
IPv6 CIDR association ID
–
Route table
rtb-049d29bfb0929dab4 | terraform-public-route-table
Auto-assign IPv6 address
No
IPv4 CIDR reservations
–
Resource name DNS A record
Disabled
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us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#InternetGateway:internetGatewayId=igw-0f3691eb7905c0398
Search
United States (N. Virginia)
VPC > Internet gateways > igw-0f3691eb7905c0398
VPC dashboard
Filter by VPC
Choose a VPC
AWS Global View
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
Carrier gateways
DHCP option sets
Elastic IPs
Managed prefix lists
igw-0f3691eb7905c0398 / terraform-igw
Actions
Details Info
Internet gateway ID
igw-0f3691eb7905c0398
State
Attached
VPC ID
vpc-0a3f8b7b22530e4f8 | terraform-vpc
Owner
281712468655
Tags (1)
Manage tags
Filter tags
Key
Value
Name
terraform-igw
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4). Provision EC2 for 3 different environments (Dev, Staging, and Prod) using Terraform workspaces.

Objective

To provision separate EC2 instances for Dev, Staging, and Prod environments using Terraform workspaces and environment-specific configuration values.

Implementation steps:

1. Created a project directory named terraform_workspace.

2. Created main.tf to define the AWS provider, AMI data source, EC2 resource, and instance tags.
3. Created variables.tf to define environment-specific instance types and server names.
4. Used terraform.workspace to select configuration values for the active workspace.
5. Executed terraform init to initialize Terraform.
6. Created the Dev workspace using terraform workspace new Dev.
7. Created the Staging workspace using terraform workspace new Staging.
8. Created the Prod workspace using terraform workspace new Prod.
9. Executed terraform workspace list to verify the workspaces.
10. Selected the Dev workspace and executed terraform apply.
11. Repeated the process for the Staging and Prod workspaces.
12. Verified that each workspace had its own Terraform state and that the corresponding EC2 instance was created in AWS.

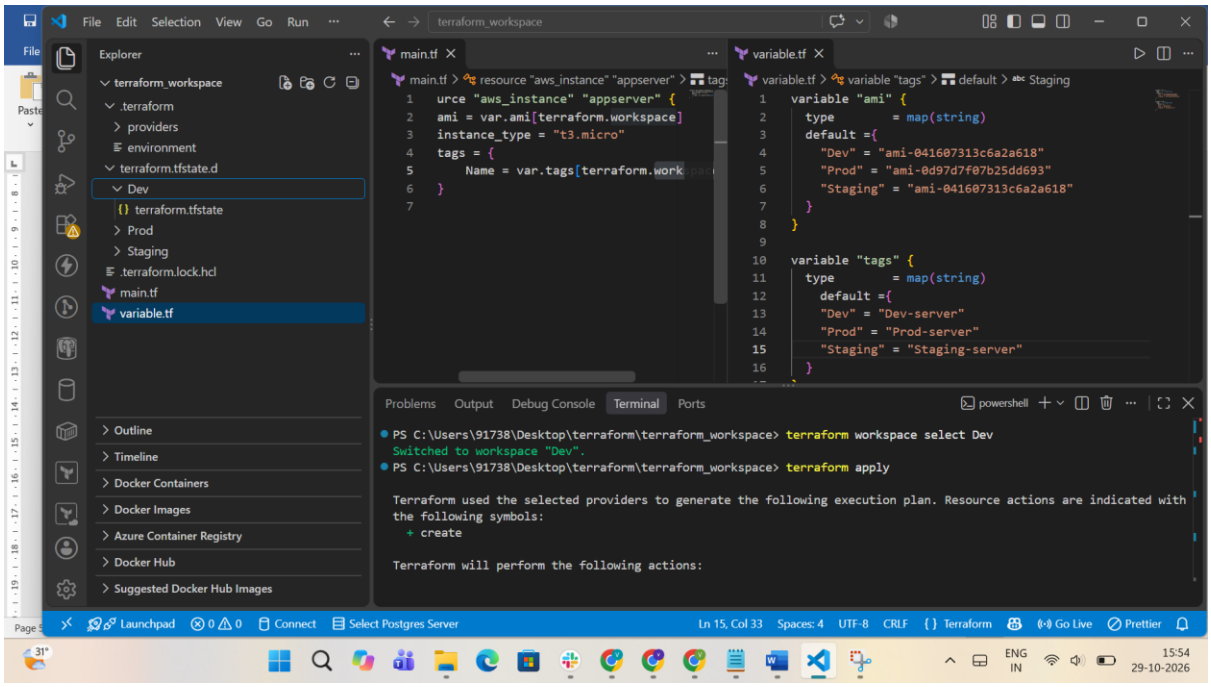
main.tf:

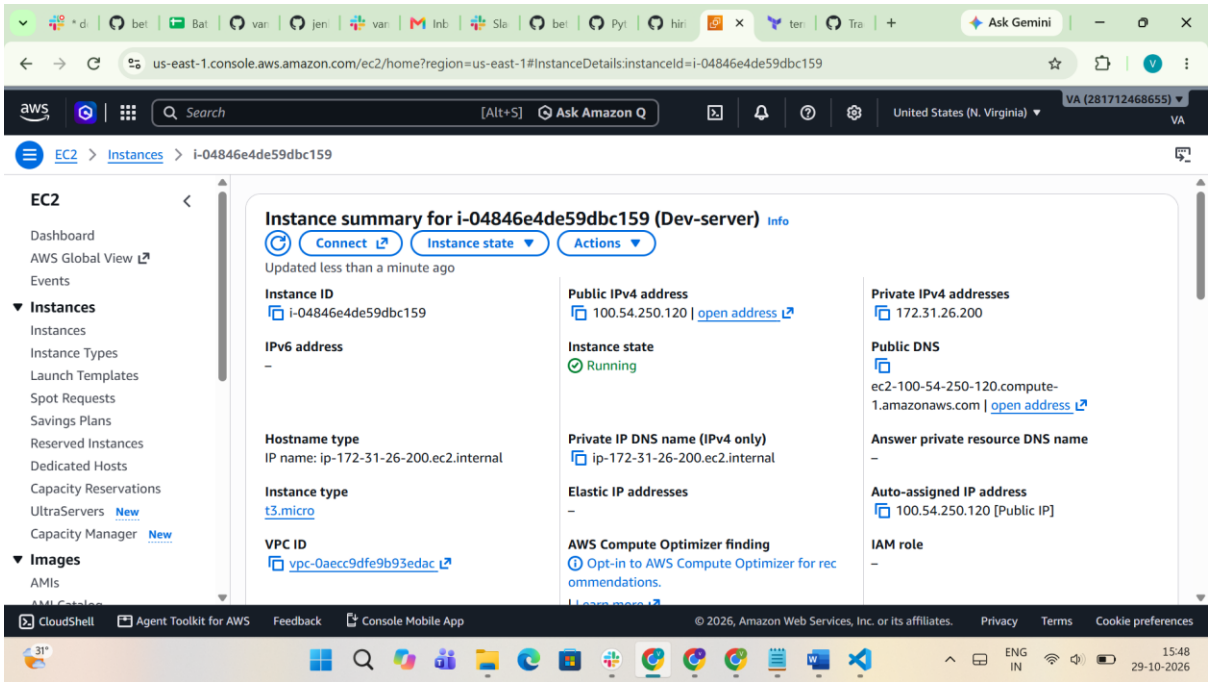
```
resource "aws_instance" "appserver" {
   ami = var.ami[terraform.workspace]
   instance_type = "t3.micro"
   tags = {
      Name = var.tags[terraform.workspace]
   }
}
```

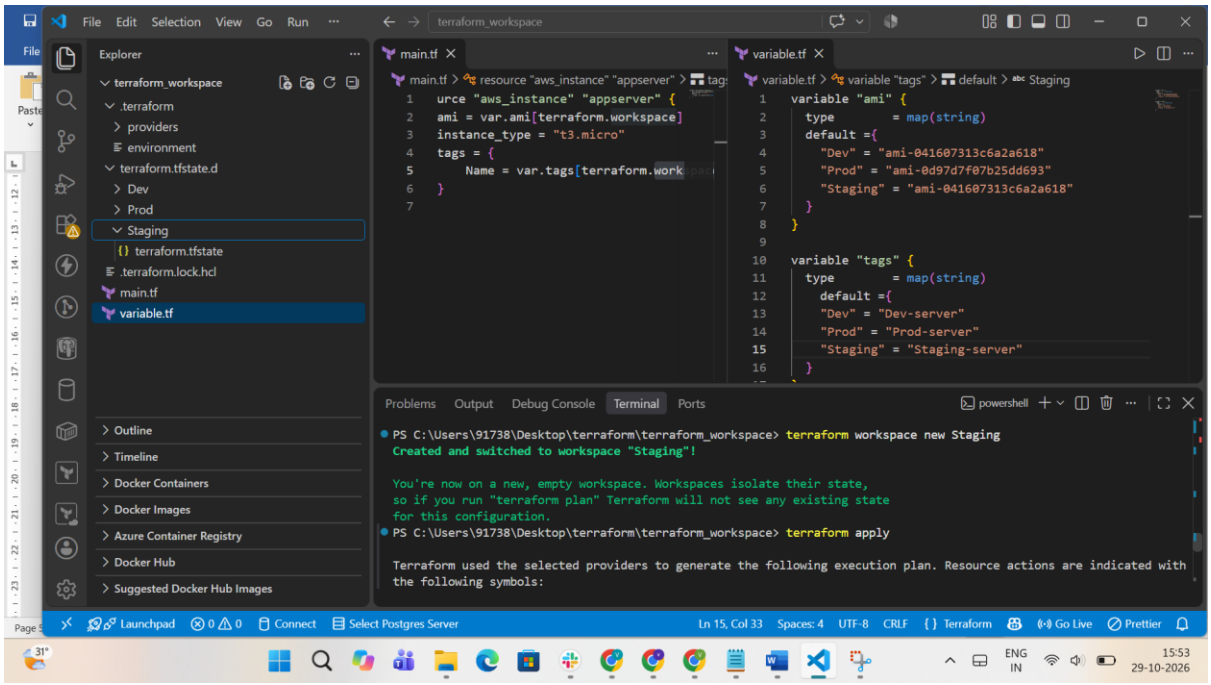
Variable.tf:

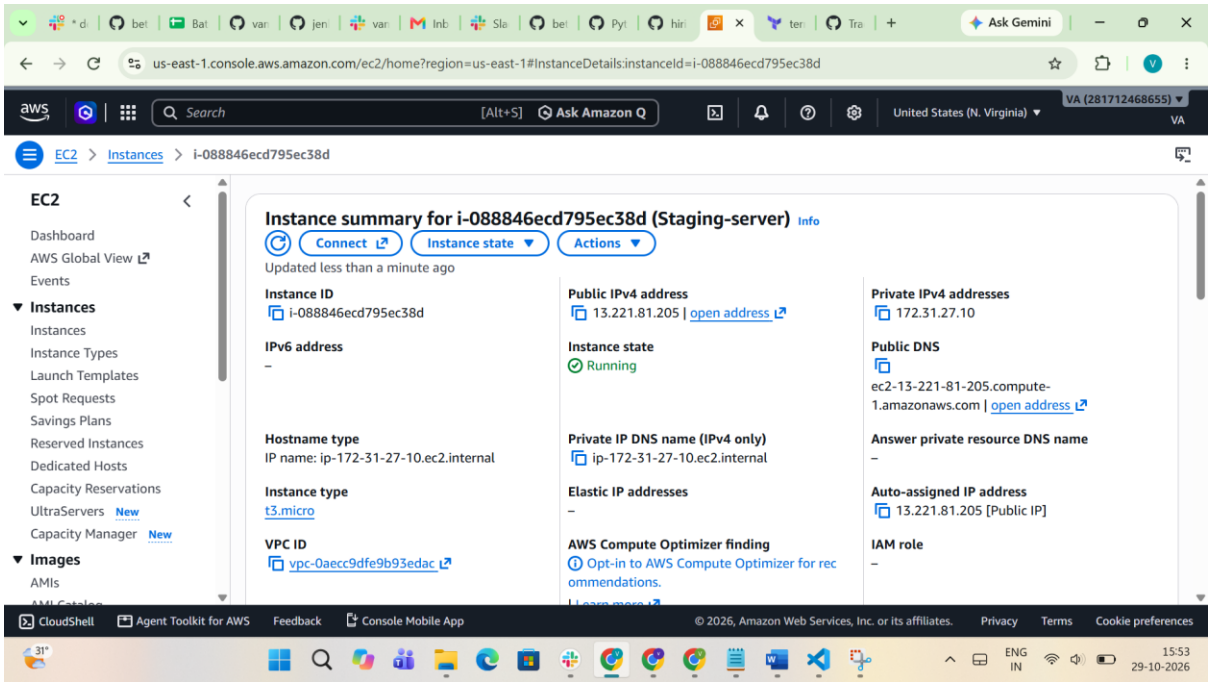
```
variable "ami" {
 type       = map(string)
 default = {
   "Dev" = "ami-041607313c6a2a618"
   "Prod" = "ami-0d97d7f07b25dd693"
   "Staging" = "ami-041607313c6a2a618"
 }
}
```

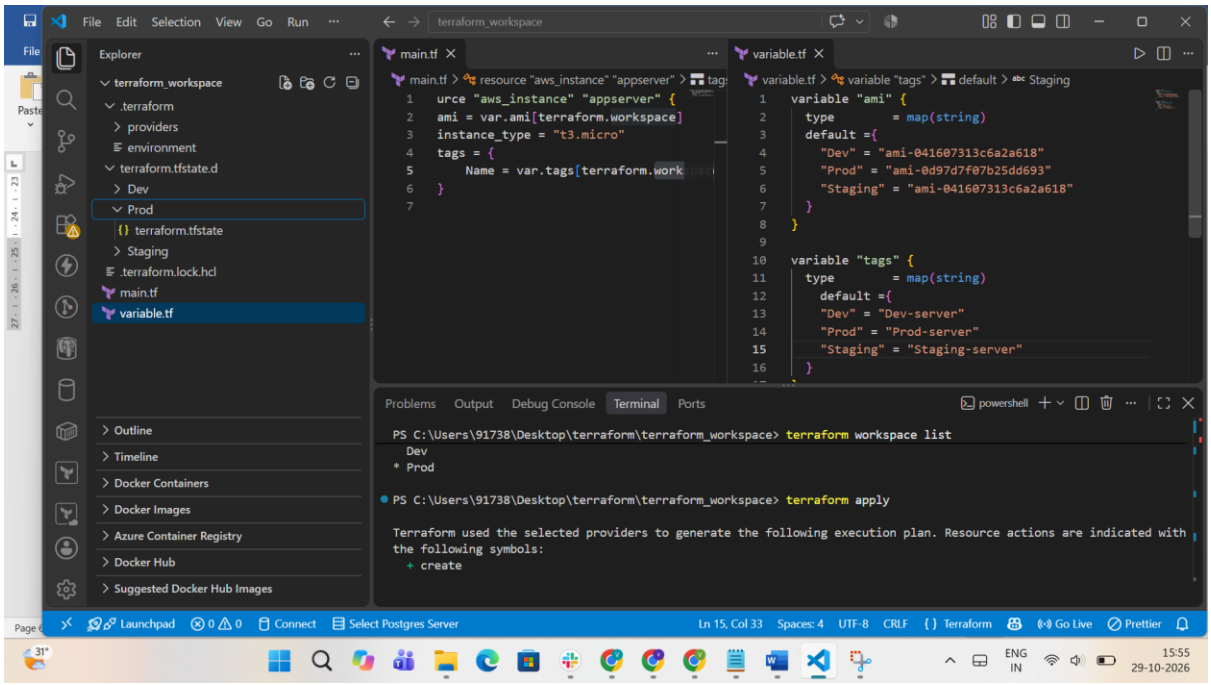
```
variable "tags" {
  type        = map(string)
    default = {
    "Dev" = "Dev-server"
    "Prod" = "Prod-server"
    "Staging" = "Staging-server"
  }
}
```

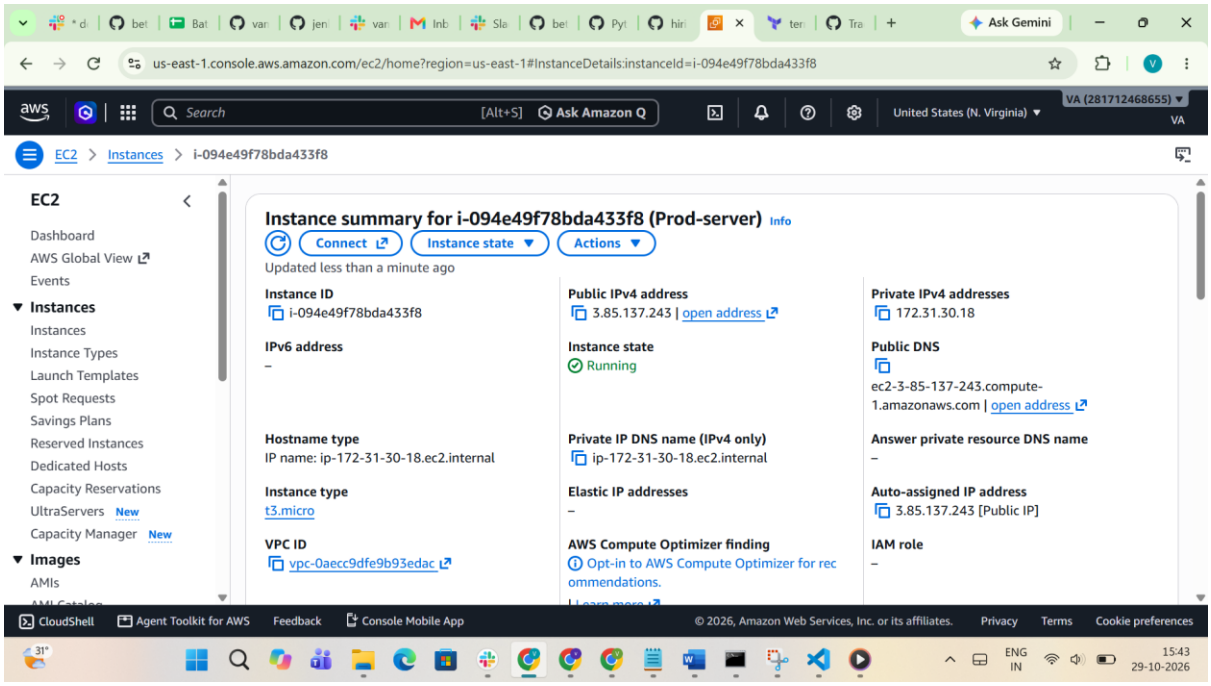



us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#InstanceDetails:instanceId=i-04846e4de59dbc159
Search
[Alt+S]
Ask Amazon Q
United States (N. Virginia)
VA (281712468655)
EC2 > Instances > i-04846e4de59dbc159
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
Instance summary for i-04846e4de59dbc159 (Dev-server) Info
Connect
Instance state
Actions
Updated less than a minute ago
Instance ID
i-04846e4de59dbc159
IPv6 address
–
Hostname type
IP name: ip-172-31-26-200.ec2.internal
Instance type
t3.micro
VPC ID
vpc-0aecc9dfe9b93edac
Public IPv4 address
100.54.250.120 | open address
Instance state
Running
Private IP DNS name (IPv4 only)
ip-172-31-26-200.ec2.internal
Elastic IP addresses
–
AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations.
Private IPv4 addresses
172.31.26.200
Public DNS
ec2-100-54-250-120.compute-1.amazonaws.com | open address
Answer private resource DNS name
–
Auto-assigned IP address
100.54.250.120 [Public IP]
IAM role
–
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File Edit Selection View Go Run
terraform_workspace
Explorer
terraform_workspace
.terraform
providers
environment
terraform.tfstate.d
Dev
Prod
Staging
terraform.tfstate
.terraform.lock.hcl
main.tf
variable.tf
main.tf
resource "aws_instance" "appserver"
urce "aws_instance" "appserver" {
ami = var.ami[terraform.workspace]
instance_type = "t3.micro"
tags = {
Name = var.tags[terraform.work
}
variable.tf
variable "tags"
default
Staging
variable "ami" {
type = map(string)
default ={
"Dev" = "ami-041607313c6a2a618"
"Prod" = "ami-0d97d7f07b25dd693"
"Staging" = "ami-041607313c6a2a618"
}
}
variable "tags" {
type = map(string)
default ={
"Dev" = "Dev-server"
"Prod" = "Prod-server"
"Staging" = "Staging-server"
}
Problems
Output
Debug Console
Terminal
Ports
powershell
PS C:\Users\91738\Desktop\terraform\terraform_workspace> terraform workspace new Staging
Created and switched to workspace "Staging"!
You're now on a new, empty workspace. Workspaces isolate their state,
so if you run "terraform plan" Terraform will not see any existing state
for this configuration.
PS C:\Users\91738\Desktop\terraform\terraform_workspace> terraform apply
Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with
the following symbols:
Outline
Timeline
Docker Containers
Docker Images
Azure Container Registry
Docker Hub
Suggested Docker Hub Images
Launchpad
Connect
Select Postgres Server
Ln 15, Col 33
Spaces: 4
UTF-8
CRLF
Terraform
Go Live
Prettier
15:53
29-10-2026


us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#InstanceDetails:instanceId=i-088846ecd795ec38d
Search
[Alt+S]
Ask Amazon Q
United States (N. Virginia)
VA (281712468655)
EC2 > Instances > i-088846ecd795ec38d
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
Instance summary for i-088846ecd795ec38d (Staging-server) Info
Connect
Instance state
Actions
Updated less than a minute ago
Instance ID
i-088846ecd795ec38d
Public IPv4 address
13.221.81.205 | open address
Private IPv4 addresses
172.31.27.10
IPv6 address
–
Instance state
Running
Public DNS
ec2-13-221-81-205.compute-1.amazonaws.com | open address
Hostname type
IP name: ip-172-31-27-10.ec2.internal
Private IP DNS name (IPv4 only)
ip-172-31-27-10.ec2.internal
Answer private resource DNS name
–
Instance type
t3.micro
Elastic IP addresses
–
Auto-assigned IP address
13.221.81.205 [Public IP]
VPC ID
vpc-0aecc9dfe9b93edac
AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations.
IAM role
–
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terraform_workspace
Explorer
terraform_workspace
.terraform
providers
environment
terraform.tfstate.d
Dev
Prod
terraform.tfstate
Staging
.terraform.lock.hcl
main.tf
variable.tf
main.tf
variable.tf
Problems
Output
Debug Console
Terminal
Ports
PS C:\Users\91738\Desktop\terraform\terraform_workspace> terraform workspace list
Dev
* Prod
PS C:\Users\91738\Desktop\terraform\terraform_workspace> terraform apply
Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create
Launchpad
Connect
Select Postgres Server
Ln 15, Col 33
Spaces: 4
UTF-8
CRLF
Terraform
Go Live
Prettier
15:55
29-10-2026



Instance state = running | Clear filters | 1

Name	Instance ID	Instance state	Instance type	Status check	Avail
Dev-server	i-04846e4de59dbc159	Running	t3.micro	3/3 checks passec	us-e
Staging-server	i-088846ecd795ec38d	Running	t3.micro	3/3 checks passec	us-e
Prod-server	i-094e49f78bda433f8	Running	t3.micro	3/3 checks passec	us-e